(Please write your Enrollment Number) Enrollment No. 02701022022

**End-Term Examination**
**(CBCS)(SUBJECTIVE TYPE)(OffLine)**
**Course Name: B.Tech ECE, Semester:3rd**
**(November-December, 2023)**

| Subject Code: BEC-207 | Subject: Digital Electronics |
|---|---|

Time :3 Hours Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

| | | | |
|---|---|---|---|
| Q1 | | (2.5*8 =20) | |
| | (a) Design a Master Slave JK flip- flop with 9 NAND gates. | | |
| | (b) Excess – 3 code is a self-complementary code, Justify | | |
| | (c) Simplify: $Y=\bar{A}C\,(\overline{\bar{A}BD}) + \bar{A}B\bar{C}\bar{D} + A\bar{B}C$ | | |
| | (d) Convert the following expression in canonical POS form. $Y = (A + B)(\bar{B} + C)$ | | |
| | (e) Differentiate between ROM, PAL and PLA with help of internal circuitry. | | |
| | (f) State the Excitation Table for JK flip-flop and T flip-flop | | |
| | (g) Draw the logic diagram of 2-to-4 line decoder using NOR gates. Draw the truth table. | | |
| | (h) Compare between the CMOS and TTL logic families | | |
| | | | |
| | **UNIT-I** | | |
| Q2 | Use K-Maps to minimize and express the function in SOP and POS form. Implement both expressions using NAND gates only. $F = \sum m\,(0, 1, 2, 6, 7, 10, 12, 15) + \sum d\,(3, 8, 13, 14)$ | (10) | |
| Q3 | (a) The 74154 is a 4-to-16 decoder. The outputs are active low and there are two gate control signals, $\overline{G1}$ and $\overline{G2}$. The decoder is enabled when $\overline{G1}$ and $\overline{G2}$ are both equal to 0. Realize the following functions using 74154 and logic gates $F_1\,(W, X, Y, Z) = \sum m\,(1, 9, 12, 15)$ $F_2\,(w, X, Y, Z) = \sum m\,(0, 1, 2, 3, 4, 5, 7, 8, 10, 11, 12, 13)$ (b) An equality detector gives the output Y = 1, if both the inputs A and B are either 1 or 0 (i) Construct the truth table (ii) Write the Boolean expression of Y (iii) Implement the circuit using NAND gates only | (5+5) | |
| | **UNIT-II** | | |
| Q4 | (a) Construct a 3-bit Ring Counter using the JK flip flops. Show the working with the waveforms w.r.t clock pulses. (b) With the help of appropriate waveforms, explain the functioning of MOD-8 Down Ripple Counter using negative edge triggered flip flops. | (5+5) | |

| | | | |
|---|---|---|---|
| Q5 | (a) Determine the output state of a 4-bit SIPO shift register after 3 clock pulses if the serial in terminal is held HIGH and the initial contents of the shift register is 1001.<br>(b) Design a self-correcting synchronous counter with the following binary sequence: 0, 1, 3, 5, 7 and repeat using T flip-flops. | (4+6) | |
| | **UNIT-III** | | |
| Q6 | (a) What is a flip flop? How is an SR flip flop realized employing<br>(i) NOR gates<br>(ii) NAND gates<br>(b) Implement an 8:1 MUX using two 4:1 MUX(s) use block diagrams. | (6+4) | |
| Q7 | (a) A 5-bit DAC produces $V_0$ = 1.4 V for a digital input of 00111. Find<br>(i) its % resolution,<br>(ii) step size ($\Delta V$),<br>(iii) $V_0$ for digital input of 11000 and<br>(iv) full scale analog voltage<br>(b)State the various performance characteristics of ADC. | (8+2) | |
| | **UNIT-IV** | | |
| Q8 | A combinational circuit is defined by the following function.<br>$F_1 (ABC) = \sum m (4, 5, 7)$<br>$F_2 (ABC) = \sum m (3, 5, 7)$<br>Implement this circuit with a PLA having 3 inputs, 3 product terms and 2 outputs. | (10) | |
| Q9 | Write a short Note on<br>(i) Content Addressable Memory<br>(ii) Dual-Slope ADC **OR** Successive-Approximation ADC | (5+5) | |